

Uncertainty Sets for Robust Process Rewards

Updated Lit Review, Foundations, and Approaches

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Abstract

We would like to investigate whether an uncertainty set around a learned process reward can reduce reward hacking under inference-time search or reinforcement learning. The challenge is to cover relevant reward errors after optimization while retaining useful discrimination between reasoning paths. These notes distinguish established findings from proposed methods, identify close prior work, explain the necessary foundations, and specify possible empirical and theoretical approaches.

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Part I

Literature Review

1. Research question and search scope

Question. Can a calibrated, structured uncertainty set around a process reward model (PRM) reduce the gap between optimized proxy reward and genuine reasoning quality? The comparison is with strong existing pessimistic methods at matched compute, annotation budget, and policy movement.

We re-searched literature on process supervision, reward hacking, reward confidence sets, pessimistic RLHF, distributional PRMs, selection-conditional conformal prediction, and robust optimization. Primary sources are linked in the references. Recent searches include September 2026 work and years distinguish initial preprints from verified conference publication where available.

1.1. What is already known, and what might be new

Static PRM accuracy and resistance to optimization are different properties. A policy or search procedure selects favorable prediction errors. Existing defenses include better supervision, reward shaping, adversarial examples, ensembles, confidence regions, and distributional rewards. Uncertainty around rewards and pessimistic optimization are established ideas. The potential contribution lies in the *coverage target, geometry, and optimization regime* for process rewards.

Four close directions must shape our claims:

1. AdvPO already constructs reward confidence regions from last-layer features and optimizes pessimistically [22].
2. PRM calibration and distributional PRMs already produce prefix-level success estimates and uncertainty bounds [24, 25].
3. Ensembles and reward-distribution DRO already give conservative effective rewards [21, 27, 23].
4. Selection-conditional conformal inference already addresses coverage after data-dependent selection [33]. Applying it to PRMs is an adaptation.

2. Process supervision: what is being estimated?

Lightman et al., Let’s Verify Step by Step (2023). Human labels supervise intermediate mathematical reasoning; PRM800K makes this supervision accessible. Process-supervised reward models outperform outcome supervision in the paper’s candidate-selection experiments, and active learning improves annotation efficiency [1]. This establishes the value of intermediate checks, and not immunity to an optimizing policy. PRM800K provides training data and a step-correctness target. A best-of- N result should not be interpreted as an RL robustness guarantee.

Wang et al., Math-Shepherd (2023 preprint; ACL 2024). Automatically generated process supervision supports reranking and stepwise PPO, reducing dependence on human labels [2]. Completion-based labels must be interpreted carefully as continuation success depends on the

generator. A confidence interval around that quantity need not cover intrinsic logical correctness. The uncertainty target must be fixed before choosing data.

Zhang et al., The Lessons of Developing PRMs (2025). The paper contrasts Monte Carlo labels, LLM judgments, and human labels and introduces consensus filtering. A correct answer can conceal flawed reasoning, and a verifier optimized for best-of- N can behave like an outcome verifier [3]. This motivates separate final-answer and process-error metrics. Qwen PRMs are practical baselines, but a minimum score is not itself calibrated uncertainty.

Xie et al., Noise-aware Learning (2026); Ding et al., SCAN (2025). Noise-aware learning uses reflection detection and iterative label refinement; SCAN combines selective Monte Carlo annotation with denoising [4, 5]. Both address corrupted synthetic supervision. Distinguish finite-rollout noise, completer dependence, and reward-model uncertainty. Better labels may improve the tube’s center without characterizing coverage under optimization. Train competing uncertainty methods on the same corrected labels.

Pala et al., Error Typing for Smarter Rewards (2025). PathFinder-PRM separates mathematical and consistency error prediction from step-correctness scoring [6]. Its reported 400K training samples should not be called 400K complete trajectories without checking the data unit. Error categories offer uncertainty features and stratified coverage diagnostics. Richer supervision is distinct from an explicit set and from resistance to an optimizing policy.

2.1. Very recent developments: error location and recovery

Wang et al., Hierarchical Multi-Step Reward Models (2025; revised 2026). HRM learns from individual and consecutive reasoning steps, while hierarchical node compression augments tree-generated supervision [49]. The method motivates modeling multi-step coherence and recovery. This is not evidence that ordinary prefix-conditioned PRMs ignore all history. A structured tube should be compared with improving the reward representation itself, especially when its claimed benefit concerns inter-step dependence.

Han et al., Cliff (September 2026 preprint). An off-the-shelf teacher locates the first mistake; Cliff assigns different advantages before and after that boundary [7]. This is a recent credit-assignment comparator and suggests uncertainty over error locations. Its effectiveness depends on teacher judgments. First-error localization does not automatically resolve self-correction or teacher exploitation.

Gan et al., Reasoning State Propagation (September 30, 2026 preprint). RSP models transitions between valid and invalid reasoning states, including break and repair probabilities, linking process labels to outcomes [8]. A weakest-step objective treats errors as irreversible; reasoning may recover. An extension could put uncertainty around break/repair transitions, requiring its own novelty comparison and more complex inference.

Fan et al., TIPS (September 29, 2026 preprint). Outcome-only RL trains a generative verifier to emit reasoning, step labels, and an outcome label [9]. This is relevant to supervision costs and generative-PRM baselines. Outcome rewards alone do not certify intermediate labels, so audit them separately. These late preprints were screened at abstract level; their gains are not independently reproduced evidence.

3. How process rewards are hacked

Cheng et al., Stop Summation / PURE (NeurIPS 2025). PURE uses minimum-based credit assignment to reduce accumulation of favorable step errors [10]. It is a required baseline: an apparent uncertainty benefit could actually result from aggregation. Reported mitigation is not a universal guarantee against degenerate outputs. Test tubes with aggregation fixed, then cross methods with sum, mean, minimum, and product.

Zheng et al., CoLD (2025; revised May 2026). CoLD targets length bias through a length adjustment, a learned bias estimator, and joint training informed by counterfactual reasoning [11]. Length and paraphrase transformations can probe our tube and motivate bias directions. A fitted estimator is not an identified causal effect without assumptions.

Tiwari et al., Reward Under Attack (2026). Static perturbations, adversarial optimization, and RL expose PRM weaknesses. In the studied AIME setup, the authors report near-perfect PRM scores while answer accuracy stays below 4% [12]. PRM-BiasBench and released code are useful resources. This is evidence about specific models and settings, not every PRM. Attack our robust score adaptively, not only the original reward.

Juneja et al., Adversarial Training for PRMs (ICML 2026). APRM co-trains an error generator and verifier to expose new failures [13]. It is a strong hard-negative comparator. Its oracle and game formulation require assumptions: own-player strong concavity alone does not imply strong monotonicity in general-sum games, and neural PPO parameters do not automatically inherit policy-space concavity. We use the empirical method without importing a blanket neural-training convergence guarantee.

Pathmanathan and Huang, REFORM (2025 preprint; ACL 2026). Reward-guided controlled decoding discovers falsely scored responses and uses them to repair preference reward models [14]. This motivates acquiring labels at optimizer-discovered errors. Compare tube construction against spending the same labels on hard-negative retraining; otherwise improved data alone could explain a gain. Its preference setting differs from process verification.

Wallace et al., Universal Adversarial Triggers (2019). Optimized token sequences expose learned NLP vulnerabilities [15]. They motivate suffix and discrete-token attacks, but do not establish transfer to a particular PRM. Distinguish content attacks from parser or delimiter exploits.

4. Reward hacking theory and optimization pressure

Skalse et al., Defining and Characterizing Reward Hacking (2022). Hackability is defined through policy-order reversals between proxy and true returns. Rich policy classes severely restrict nontrivial proxies that never reverse preferences [16]. This does not prove every training run hacks. It motivates conditional guarantees over restricted or reachable policy classes rather than globally perfect alignment.

Gao et al., Scaling Laws for Reward Model Overoptimization (2022). Experiments with a stronger gold reward model show proxy scores increasing after gold scores deteriorate [17]. Use optimization-pressure curves, not one checkpoint. The gold model remains a proxy for human intent; exact mathematics and independent step audits are preferable where available. Empirical scaling laws are not universal laws for PRM reasoning.

Laidlaw et al., Correlated Proxies (2024 preprint; ICLR 2025). Proxy–true correlation under a reference policy supports reward-hacking theory and occupancy-measure regularization [18]. Reference-distribution correlation is distinct from pointwise validity in new states. For our project, ask how a reward-set assumption yields a useful bound under changing reasoning prefix distributions.

Liu, Sun, and Zheng, Robust Optimization with Correlated Proxies (ICLR 2026). The paper optimizes against worst-case rewards constrained by proxy correlation and studies known linear reward features [19]. Neither correlation sets nor generic max–min hacking defenses are new. A PRM contribution needs process-specific geometry, construction from supervision, or guarantees and evidence under selection and on-policy shifts.

Liu et al., Robust General Utility for RL (August 2026 preprint). Utility functionals of occupancy are uncertain; algorithms address different utility regimes [20]. This is relevant to nonlinear PRM aggregation. It was screened at abstract level: inspect the full assumptions before using its stationarity results. Generic robust utility optimization is already prior work.

5. Closest prior work on uncertainty and pessimism

Coste et al., Reward Model Ensembles (2023 preprint; revised 2024). Mean, worst-case, and uncertainty-weighted ensembles are tested under best-of- N and PPO, including noisy labels [21]. Pessimism reduces overoptimization in the tested synthetic reward setting. Shared data or representations can make members confidently wrong together. Include minimum-member and variance-penalty baselines; simply using a lower ensemble score is not a new contribution.

Zhang et al., AdvPO (2024). AdvPO derives uncertainty from last-layer embeddings, constructs a confidence region, and performs adversarial policy optimization [22]. This is the closest predecessor to a feature-based ellipsoidal tube. Theory assumes reward approximation and feature structure; its practical radius is tuned. Potential advances are process-level joint coverage, calibrated misspecification, and selected-trajectory audits. Transferring its geometry to a PRM alone is a limited extension.

Yan et al., Reward-Robust RLHF (2024). Bayesian reward ensembles and a nominal-pessimistic tradeoff already model uncertain rewards [23]. An ensemble distribution is a modeling device, not automatic coverage of truth. Check coverage, sharpness, and systematic bias independently.

Park et al., Know What You Don’t Know (2025). Quantile regression calibrates PRM-derived continuation-success probabilities for a specified generator. Bounds support adaptive inference budgets [24]. This is directly relevant, but its target differs from intrinsic step correctness. Calibration and efficiency do not automatically give joint coverage across selected candidates or an RL hacking guarantee. A faithful implementation is a required baseline.

Ma et al., Distributional PRMs via Conditional Optimal Transport (May 2026). Conditional OT learns a monotone conditional quantile map using PRM features and rollout-success targets [25]. It offers coherent quantiles and flexible confidence levels. Quantile monotonicity differs from finite-sample coverage after selection; its target also changes with the completer. Our advance must be more specific than learning reward distributions or confidence bounds.

Yu et al., From Curiosity to Caution (ICLR 2026). Caution penalizes prediction-error novelty to mitigate best-of- N reward hacking, with empirical and simplified theoretical analysis [26]. Novelty is not logical error: unusual correct reasoning may be penalized while familiar wrong reasoning evades it. Compare calibrated residual uncertainty against novelty penalties at matched scoring cost.

Hahami et al., A Unifying Lens on Reward Uncertainty (June 2026). A distributional reward yields an entropic pessimistic score; mean, minimum, and variance penalties arise as limits or approximations [27]. Reward robustness and policy KL have distinct roles. Entropic aggregation is already prior work. Joint process-step distributions or policy-dependent coverage could extend it, provided model misspecification is addressed.

Zhou et al., PRISM (ACL 2026); Duan et al., BNRM (2026 preprint). PRISM uses Gaussian mixtures to represent preference heterogeneity and uncertainty; BNRM uses Bayesian nonnegative latent factors [28, 29]. These concern preference rewards rather than certified process correctness. Rich output distributions need not cover reasoning failures. Separate preference diversity, annotation variability, and epistemic errors.

6. Statistical tools for constructing a meaningful set

Lakshminarayanan et al., Deep Ensembles (2017). Independently trained predictors provide practical uncertainty estimates [30]. Seeds, data resamples, and backbones give different diversity. Measure systematic common errors. Dispersion is an input to calibration, not a confidence-set theorem by itself.

Abbasi-Yadkori et al., Linear Stochastic Bandits (2011). Self-normalized martingale bounds support confidence ellipsoids for adaptive features under a linear conditional mean and noise assumptions [31]. Changing neural features, logistic labels, or corrupted annotations require new arguments. A frozen-feature regression setting isolates these assumptions.

Romano et al., Conformalized Quantile Regression (2019). Conformal calibration corrects learned quantiles to give marginal prediction coverage under exchangeability [32]. It covers future observed targets, not automatically conditional means or whole reward functions. Calibrate by problem rather than treating correlated steps as independent examples.

Jin and Ren, Confidence on the Focal (2024 preprint; JRSS B 2025). Selection-conditional inference covers focal test units chosen by data-dependent rules, including optimization-based selection [33]. This is close to PRM best-of- N . An adaptation must check rule symmetry, target, and computational cost. It does not automatically cover an indefinitely co-adapting policy and verifier.

Tibshirani et al., Covariate Shift (2019); Barber et al., Beyond Exchangeability (2022). Weighted and nonexchangeable conformal methods address specified shifts [34, 35]. Density ratios, support overlap, and a stable conditional label mechanism are substantive assumptions. Estimated or clipped weights do not preserve exact coverage automatically. Changing the completer changes a rollout-success target, so that is not mere covariate shift.

Angelopoulos et al., Conformal Risk Control (2022). Calibration extends to expected monotone losses [36]. This motivates controlling false acceptance of bad steps. The theorem must match the loss and monotonicity: selecting a different trajectory at each threshold can break monotonicity. Start with thresholded accept/reject rules on fixed scores.

Bao et al., Model Selection for Conformalized Robust Optimization (2025). Prediction sets, model selection, and downstream robust decision quality are connected [37]. A narrow valid interval is not necessarily best for a decision. Decision-aware calibration is prior work; report accuracy and regret alongside width.

6.1. Late statistical results to monitor

Two 2026 preprints were screened: a testing view of conformal universality and impossibility [38], and finite-sample CQR bounds including fixed-score calibration under known shift [39]. They may inform coverage–efficiency lower bounds. Full assumptions need inspection; no theorem below is attributed to these abstracts.

7. Robust optimization and reward shaping

Duchi and Namkoong, Uniform Performance via DRO (2018 preprint). Distributional ambiguity emphasizes difficult examples and tail behavior, with statistical tradeoffs [40]. Robust PRM training can reweight ordinary and discovered failures, but empirical reweighting cannot create unseen reasoning behaviors. A set of data distributions differs from a set of reward functions.

Iyengar, Robust Dynamic Programming (2005). Rectangular transition uncertainty supports robust dynamic programming [41]. The structural lesson transfers: independent local adversarial choices and one globally consistent model are different problems. Text transitions may be known while PRM reward uncertainty is coupled. Do not assume a robust Bellman recursion for an arbitrary nonrectangular tube.

Ng, Harada, and Russell, Policy Invariance (1999). Potential-based shaping preserves policy preferences under appropriate boundary conditions [42]. Where outcome verification is reliable, a PRM can assist learning as a potential. This differs from replacing rewards with pessimistic estimates. It does not remove verifier errors or promise faster learning.

8. Evaluation resources and evidence gaps

ProcessBench tests earliest-error identification on expert-annotated mathematical solutions [43]; PRMBench tests dimensions including simplicity, soundness, and sensitivity [44]; Socratic-PRMBench adds reasoning-pattern diagnostics [45]. PRM-BiasBench supplies controlled perturbations [12]. Static performance cannot replace optimization stress tests.

Family	Established capability	What our study must resolve
Process supervision	Intermediate verification	What quantity the tube covers
Label denoising	Better synthetic labels	Coverage after optimization
PURE / shaping	Better credit assignment or outcome invariance	Tube benefits with aggregation held fixed
Ensembles / AdvPO	Pessimistic reward optimization	Shared bias and joint process uncertainty
Quantile / OT PRMs	Prefix distributions and bounds	Selection coverage and completer dependence
Conformal / JOMI	Guarantees for specified sampling rules	Feasible PRM adaptation and adaptive RL assumptions
Correlated-proxy robustness	Structured worst-case rewards	Construction from process supervision

8.1. A defensible candidate contribution

Optimization-aware uncertainty for process rewards: compare pointwise intervals with structured or problem-level tubes; measure coverage after selection; determine whether coverage improvements reduce hacking at matched optimization pressure. Novelty remains provisional. An empirical paper could expose systematic post-selection undercoverage and a practical repair. A theory paper could characterize safe improvement and failure of marginal intervals. A combined paper would connect a restricted theorem to measured violations in real PRMs. No publication outcome is assumed.

8.2. Reading order

Read Part II, then Lightman and the Qwen lessons for targets; PURE and Reward Under Attack for failures; AdvPO, Park, conditional OT, Caution, and reward uncertainty unification for close

methods; JOMI and correlated-proxy robustness for statistical distinctions. Part III develops concrete designs.

Part II

Notes: Foundations for the Project

9. Notation and the objectives we must keep separate

Symbol	Meaning
q	Problem or prompt drawn from a task distribution
s_t, h_t	Reasoning step and prefix $h_t = (q, s_1, \dots, s_t)$
τ, T	Complete trajectory and number of reasoning steps
π_θ, π_0	Policy being optimized and fixed reference policy
z_t	State before action a_t ; in text, the prompt and current prefix
c_t, v_t	Local step-correctness and cumulative prefix-validity labels
$p_{\pi_c}(h_t)$	Probability a fixed completer π_c finishes correctly
$\hat{r}_\phi, r^*$	Learned process reward and specified target process reward
G	Explicit step-to-trajectory aggregation rule
$\hat{R}, R^*$	Proxy trajectory reward and intended task utility
$\mathcal{U}, L, U$	Reward uncertainty set and its lower and upper envelopes
b, α, δ	Width; prediction miscoverage level; confidence-set failure probability
ρ, β	Ambiguity-set radius and policy KL regularization coefficient

Critical distinction. In general,

$$c_t \neq v_t \neq p_{\pi_c}(h_t), \quad G(r^*(h_1), \dots, r^*(h_T)) \neq R^*(\tau).$$

A valid algebraic manipulation can follow an earlier false assumption; a wrong step can later be repaired; a correct prefix can be difficult for a weak model to complete. Even perfectly estimating step correctness does not automatically make an arbitrary aggregation equal final-answer accuracy.

10. Reinforcement learning in the reasoning setting

10.1. MDPs, states, actions, and policies

A finite-horizon Markov decision process specifies states, actions, transition probabilities, rewards, an initial-state distribution, and a horizon. The Markov property says that, conditional on the current state and action, the next-state distribution does not additionally depend on earlier history:

$$\mathbb{P}(z_{t+1} \mid z_{0:t}, a_{0:t}) = P(z_{t+1} \mid z_t, a_t).$$

It does not mean only the immediately preceding observation matters: the state can encode the entire history. Text generation is Markov when the state includes the full prompt and generated prefix. An action can be one token, one reasoning step, or a tool call. Appending text is deterministic given the action; sampling actions from the policy is stochastic. Tool environments may also have stochastic or hidden state. A partially observable formulation is needed when observations do not determine the environment state.

A stochastic policy maps a state to an action distribution. A deterministic policy picks one action. A stationary policy depends on the current state, and a nonstationary one may also depend on time. Time can be appended to the state in a finite-horizon problem. Existence of stationary deterministic optima requires the usual MDP conditions and objective; do not assert it for arbitrary partial observability or nonlinear robust objectives.

10.2. Returns, value, and credit assignment

Use the convention that r_t is earned by action a_t :

$$G_t^{\text{sum}} = \sum_{k=t}^{T-1} \gamma^{k-t} r_k, \quad V^\pi(z) = \mathbb{E}_\pi[G_t^{\text{sum}} \mid z_t = z], \quad Q^\pi(z, a) = \mathbb{E}_\pi[G_t^{\text{sum}} \mid z_t = z, a_t = a].$$

The advantage $A^\pi(z, a) = Q^\pi(z, a) - V^\pi(z)$ measures whether an action is better than the policy's average action. For additive returns,

$$Q^\pi(z, a) = \mathbb{E}[r_t + \gamma V^\pi(z_{t+1}) \mid z_t = z, a_t = a].$$

Sparse terminal rewards make it hard to determine which earlier decisions caused success. Dense process feedback may help, but repeated erroneous rewards also accumulate. The classical Monte Carlo, temporal-difference, value-learning, and actor-critic vocabulary is useful background; these notes use a finite episodic reasoning model rather than reproduce all tabular algorithms. The textbook is a background reference [46].

10.3. Policy gradients, PPO, and GRPO

For a fixed trajectory reward, the likelihood-ratio identity gives

$$\nabla_\theta \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)] = \mathbb{E} \left[R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t \mid z_t) \right].$$

Baselines reduce variance; return-to-go or advantage estimates improve credit assignment. A critic estimates value; it is a separate source of approximation error from the reward model.

PPO commonly uses the clipped surrogate [47]

$$\mathbb{E}_t \left[\min\{\omega_t(\theta) \hat{A}_t, \text{clip}(\omega_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t\} \right], \quad \omega_t(\theta) = \frac{\pi_\theta(a_t \mid z_t)}{\pi_{\text{old}}(a_t \mid z_t)}.$$

Clipping limits this surrogate's incentive for large individual updates; it is not a hard guarantee that policy KL remains small. The policy can still move far from its reference through many small updates.

GRPO estimates relative advantages from groups of sampled answers, avoiding a separately learned value critic in its common formulation [48]. A simple outcome-level version uses

$$\hat{A}_j = \frac{R_j - \bar{R}}{\text{std}(R_1, \dots, R_m) + \varepsilon}.$$

Dense process rewards require a stated token/step assignment; broadcasting one trajectory advantage to every token is not stepwise process credit assignment. Group normalization can obscure raw reward scales and gives little signal when all group rewards coincide. Document the actual variant in experiments.

10.4. RLHF, RLVR, and inference-time search

RLHF commonly trains a preference reward model and optimizes against it. RLVR uses automatically checked outcomes, such as answers or test results. The checker is only as reliable as its specification and implementation. PRMs can provide dense supervision in either setting. Best-of- N changes which sampled trajectory is selected while keeping generator weights fixed:

$$\tau_{\text{sel}} = \arg \max_{j \leq N} \hat{R}(\tau_j), \quad \tau_j \sim \pi_0.$$

Beam search additionally selects prefixes and changes the effective exploration distribution. RL changes generator parameters, allowing repeated adaptation to the reward’s errors. Results in one regime need not transfer to another.

11. What a process reward model does

A discriminative PRM predicts correctness scores at step boundaries from the question and preceding reasoning. A generative PRM emits critiques or labels. Conditioning on preceding steps is common; “stepwise” does not mean the model ignores all previous context. The reward head may output a logit, a probability, or a free scalar. The domain and scale determine admissible uncertainty sets.

11.1. Labels and training

For binary labels, a standard loss is

$$\ell_{\text{BCE}}(\phi; h_t, y_t) = -y_t \log \hat{r}_\phi(h_t) - (1 - y_t) \log(1 - \hat{r}_\phi(h_t)).$$

Human annotation may stop at the first error. Later steps then have missing or censored labels; treating them all as independently labeled local errors changes the target. Automated judges introduce correlated annotation mistakes.

For K independent completions under a fixed completer π_c ,

$$\tilde{p}(h_t) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{\text{completion } k \text{ succeeds}\}, \quad \text{Var}(\tilde{p} \mid h_t) = \frac{p_{\pi_c}(1 - p_{\pi_c})}{K}.$$

A label for “at least one success” has mean $1 - (1 - p_{\pi_c})^K$, not p_{π_c} . It changes with K and can approach one for any nonzero success probability. Model repair makes completion success different from correctness of the present step. Thus Monte Carlo uncertainty and semantic label mismatch are distinct problems.

11.2. Aggregation creates additional incentives

For probability-like scores $r_t \in [0, 1]$, common choices are

$$G_{\text{sum}} = \sum_t r_t, \quad G_{\text{mean}} = T^{-1} \sum_t r_t, \quad G_{\text{min}} = \min_t r_t, \quad G_{\text{prod}} = \prod_t r_t.$$

Sum rewards repetition of easy steps. Mean can dilute a bad step with easy ones. Minimum emphasizes the weakest step but can reward short or vacuous responses and rejects repair unless repair is explicitly modeled. Product penalizes length: $0.95^{10} \approx 0.60$ and $0.95^{30} \approx 0.21$. It is a joint-correctness probability only under an appropriate probabilistic interpretation; multiplying ordinary marginal step scores does not justify an independence claim.

For bounded per-step perturbations $|r_t - \tilde{r}_t| \leq \epsilon$, sum errors can grow as $T\epsilon$, while minimum errors are bounded by ϵ . This deterministic observation concerns the aggregation target, not proof that minimum rewards optimize true mathematical accuracy. Splitting or merging steps can change all four scores. Include segmentation attacks and empty-output rules.

12. Why reward hacking occurs and why it matters

Write $\hat{R}(\tau) = R^*(\tau) + e(\tau)$ schematically. Maximizing $\hat{R}$ selects both genuine quality and positive error. Error on ordinary data is not the same as error among the candidates chosen by the optimizer.

12.1. A numerical example

Suppose a correct solution scores 0.80 while an incorrect fluent solution scores 0.95. A policy increasing the latter’s probability improves proxy reward and reduces correctness. A useful tube might give the correct solution $[0.76, 0.84]$ and the exploitable solution $[0.20, 1.00]$, making pessimistic selection prefer the correct solution. If both intervals are too narrow, or the wrong solution’s uncertainty is underestimated, this defense fails. If every interval is $[0, 1]$, the lower score gives no useful ranking.

12.2. Different mechanisms need different diagnostics

- **Selection of noise:** higher search budgets find larger positive errors even without new policy weights.
- **Distribution shift:** optimization visits reasoning prefixes absent from training, where feature–correctness associations fail.
- **Misspecification:** a scorer rewards length, style, or easy local claims instead of the intended reasoning property.
- **Objective mismatch:** perfectly estimated local rewards are summed or transformed into a trajectory objective that rewards the wrong behavior.
- **Evaluator manipulation:** delimiters, injected instructions, or checker loopholes change what the evaluator measures.

Optimization does not require conscious intent to exploit a reward. Conversely, an ordinary wrong answer is not by itself evidence of hacking. Look for reward improvement coupled to independent quality stagnation or decline and a mechanism that explains it. Hacking harms model quality, invalidates capability estimates, wastes compute, and can train persistent undesirable behavior.

13. Uncertainty, calibration, and coverage

13.1. Three kinds of uncertainty

Epistemic uncertainty concerns limited knowledge of the target reward function. **Aleatoric variability** concerns stochastic outcomes or ambiguous labels given the input. **Misspecification** means the model class or label definition systematically misses relevant truth. Ensemble variance may detect the first while missing the third. Rollout variance primarily measures outcome variability and finite-sample noise, not intrinsic logical ambiguity.

13.2. Prediction intervals are not confidence bands for a function

An interval covering a new binary label is a prediction set for Y . A confidence interval for $p(x) = \mathbb{P}(Y = 1 \mid X = x)$ covers a conditional mean. A function confidence set asserts simultaneous containment of p over a domain. They require different information and assumptions. A binary-label interval that contains both zero and one is valid but gives little reward discrimination. Quantiles of observed labels or rollout estimates do not automatically become confidence bounds on intrinsic correctness probabilities.

Calibration of a probability score μ means that among examples assigned a score near u , the event frequency is near u . This is distribution-dependent. Low aggregate expected calibration error can coexist with poor accuracy on rare high-scoring mistakes. Ranking accuracy and calibration are also different.

Marginal coverage is

$$\mathbb{P}\{Y_{\text{new}} \in C(X_{\text{new}})\} \geq 1 - \alpha.$$

Trajectory coverage contains all relevant steps. Candidate-set coverage contains all trajectories being searched. Selection coverage concerns the chosen one. Function-level coverage contains the target reward on every point of a stated domain. Repeated RL adaptation requires an event valid for all encountered queries or a sequentially justified update procedure; a static marginal claim does not provide it.

14. Robust optimization: the geometry matters

For $\hat{r} : \mathcal{H} \rightarrow [0, 1]$, a pointwise box tube is

$$\mathcal{U}_{\text{box}} = \{r : |r(h) - \hat{r}(h)| \leq b(h), h \in \mathcal{H}\}.$$

It allows each point's error to vary independently. A function set induced by one uncertain parameter w couples different prefixes. An adversary choosing one reward function for the entire policy gives

$$\max_{\pi} \inf_{r \in \mathcal{U}} \mathbb{E}_{\tau \sim \pi} [G(r(h_1), \dots, r(h_T))].$$

An adversary choosing a different model after each trajectory gives

$$\max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\inf_{r \in \mathcal{U}} G(r(h_1), \dots, r(h_T)) \right].$$

Generally the second is no larger than the first. They can coincide for certain rectangular sets but need not for coupled sets. State explicitly which is solved.

Distributional robustness instead permits an adversary to change a distribution. For robust training this might be

$$\min_{\phi} \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\ell_{\text{PRM}}(\phi; h, Y)].$$

It changes the fitted model, whereas policy pessimism changes how a fixed model is used. Statistical DRO over examples, DRO over uncertain reward values, and robust control over functions should not share one undifferentiated symbol.

15. Distribution movement and reward shaping

For $\gamma < 1$, define normalized discounted occupancy

$$d_{\pi}(z, a) = (1 - \gamma) \mathbb{E}_{\pi} \sum_{t \geq 0} \gamma^t \mathbf{1}\{z_t = z, a_t = a\}.$$

Additive expected return is $(1 - \gamma)^{-1} \langle d_{\pi}, r \rangle$. For finite-horizon reasoning with $\gamma = 1$, use the finite sum of state-action visitation measures instead. Policy action KL, trajectory KL, and occupancy divergence are different quantities. Under shared dynamics, trajectory KL can be expressed as the expected sum of conditional action KLs, but an empirically averaged token KL still needs normalization and sampling conventions.

An error bound averaged under d_{π_0} may be amplified under d_{π} . Importance weighting requires $d_{\pi} \ll d_{\pi_0}$ and a usable density ratio; theoretical support overlap can be practically useless if weights are enormous.

With a reliable base task reward, potential shaping uses

$$F_t = \gamma \Phi(z_{t+1}) - \Phi(z_t).$$

The discounted sum telescopes:

$$\sum_{t=0}^{T-1} \gamma^t F_t = -\Phi(z_0) + \gamma^T \Phi(z_T).$$

Setting the terminal potential to zero removes dependence on chosen actions [42]. Fix the potential during each evaluated episode and handle truncation explicitly. Arbitrary dense PRM bonuses, adaptive within-episode potentials, or nonzero length-dependent terminal potentials do not inherit this argument.

16. Research habits that make the results interpretable

Fix the target and label protocol before selecting an uncertainty estimator. Split data by problem, retain independent calibration and evaluation sets, and keep attack construction separate from final transfer tests. Report correctness, process validity, uncertainty coverage, computational cost, and reward curves. An uncertain reward estimator cannot establish its own success merely by giving higher scores. The detailed experiment plan and decision rules follow in Part III.

Part III

Constructing and Characterizing the Reward Tube

17. From the line metaphor to a research object

Along a trajectory, the PRM produces the line $\hat{r}(\tau) = (\hat{r}(h_1), \dots, \hat{r}(h_T))$. A tube should contain plausible target rewards, not merely decorate that line with arbitrary error bars. Define lower and upper envelopes $L(h), U(h)$, a domain $\mathcal{H}_B$ indexed by an optimization budget B , and a set of functions

$$\mathcal{U}_B = \{r : \mathcal{H}_B \rightarrow [0, 1] : L(h) \leq r(h) \leq U(h), \quad r \text{ satisfies specified coupling constraints}\}.$$

The coupling says which combinations of step errors are simultaneously plausible. Independent intervals form a box; shared parameters form a correlated tube. This distinction is the main geometric research question.

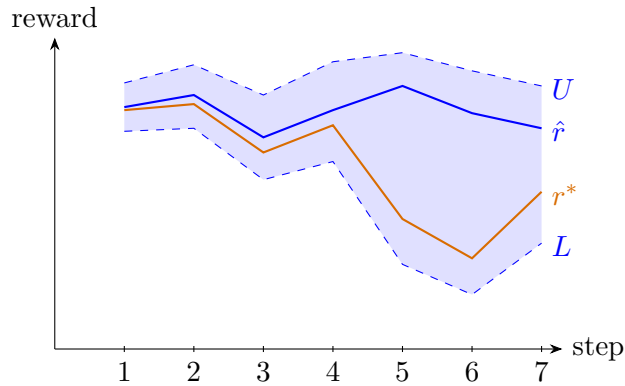


Figure 1: Illustrative tube with heterogeneous width. All values are schematic, not measured results. Drawing a target line inside the band does not establish statistical coverage or describe which correlated functions are admissible.

17.1. The design specification

Before implementing any method, specify:

1. **Target:** human/prover step validity, cumulative prefix validity, fixed-completer success probability, or task utility. We recommend intrinsic step/prefix validity as the primary PRM target and independent answer correctness as the task metric. Use a separate fixed-completer track for rollout calibration.
2. **Domain:** a finite candidate pool, trajectories from a frozen policy, a bounded policy family, or all feature vectors in a known domain.
3. **Geometry:** independent box, shared parameter ellipsoid, low-rank bias set, budgeted deviations, or a joint reward distribution.

4. **Assurance:** empirical coverage, marginal prediction coverage, candidate-set coverage, uniform function coverage, or a policy improvement bound.
5. **Use:** robust reranking, trajectory-level RL, robust policy-level improvement, or task-preserving shaping. These are different objectives.

The methods below are proposals. Their novelty and effectiveness are hypotheses.

18. Approach A: problem-level calibration of candidate tubes

Purpose. Establish a practical baseline and diagnose whether selection breaks ordinary coverage. This is an empirical route with a modest, explicit exchangeability guarantee for a frozen candidate-generation protocol.

18.1. Construction

Train a center $\mu(h) \in [0, 1]$ and positive scale $s(h) \geq s_{\min} > 0$ using only training problems. Scale candidates include ensemble disagreement, last-layer leverage, predicted residual magnitude, or quantile widths. Fix the scorer, generator, parser, and maximum budget B before calibration. For each calibration problem i , generate all B candidates and collect target labels y_{ijt} for their steps. Problems, not individual steps, are the exchangeable calibration units. Define

$$S_i^- = \max_{j \leq B, t \leq T_{ij}} \frac{\mu(h_{ijt}) - y_{ijt}}{s(h_{ijt})}, \quad S_i^+ = \max_{j \leq B, t \leq T_{ij}} \frac{y_{ijt} - \mu(h_{ijt})}{s(h_{ijt})}.$$

On m calibration problems, let $q^\pm$ be the $[(m+1)(1-\alpha_\pm)]$ th order statistic of $S_i^\pm$, augmented with $+\infty$ if the index exceeds m . Replace negative quantiles by zero to make widths nonnegative. Set $\alpha_- + \alpha_+ = \alpha$ and

$$L(h) = \max\{0, \mu(h) - q^- s(h)\}, \quad U(h) = \min\{1, \mu(h) + q^+ s(h)\}.$$

A one-sided version spends all the error budget on S^- and reports only a lower prediction bound; it must not be called a two-sided confidence tube.

Under exchangeable problem blocks, independent model fitting, and the same frozen generation/labeling protocol, split-conformal ranks and a union bound give

$$\mathbb{P}\{L(h_{jt}) \leq y_{jt} \leq U(h_{jt}) \text{ for all candidates and steps in a new block}\} \geq 1 - \alpha.$$

This adapts standard conformal reasoning [32]; it is not a new theorem. Within-block steps and candidates can be dependent. Any reranker choosing from that fixed block inherits this containment event. The probability is marginal over calibration and new problem blocks, not conditional coverage for every problem or a confidence band over the whole reward function.

18.2. The target problem, and how to handle it

If y is a binary validity label, this construction covers the *realized label*. Many honest intervals may reach zero, making lower-bound selection uninformative. That is a result to report, not something to

fix by relabeling the interval as epistemic uncertainty. Use sets of valid/invalid labels or abstention for this track; evaluate informativeness through singleton-set frequency and false acceptance.

For a success-probability track, keep completer π_c fixed and label each prefix with K independent completion outcomes. Calibration then covers future rollout estimates $\tilde{p}$, not p_{π_c} . If the new block has at most $M = BT_{\max}$ prefixes, Hoeffding and a union bound give an additional allowance

$$\eta_K = \sqrt{\frac{\log(2M/\delta_{\text{MC}})}{2K}}.$$

Expanding each interval by η_K covers the true fixed-completer success probabilities in that block with probability at least $1 - \alpha - \delta_{\text{MC}}$. Independent Bernoulli rollouts, a correct answer checker, and a fixed finite cap are assumptions. This does not convert success probabilities into intrinsic step correctness.

18.3. Using the tube

For the rectangular set and a coordinatewise nondecreasing aggregation G , rerank using

$$G(L(h_1), \dots, L(h_T)).$$

Repeat for nominal, calibrated-center, and uncertainty-penalized baselines with the same G . A candidate-wise lower reward may be overly conservative; it should be compared with a shared-function set in Approach B.

18.4. Tests, limitations, and novelty

Sweep nested fixed pools with $N \leq B$. Compare stepwise calibration, trajectory-block calibration, full candidate-block calibration, a union-bound correction, and a selection-conditional/JOMI adaptation [33]. Calibrating the selected candidate directly requires the selection protocol to be fixed without circular reuse of its calibration labels.

Problem blocks from an RL-updated generator or a beam search influenced by the new calibrated reward do not automatically match the frozen-pool guarantee. Block maxima can be too conservative and annotation cost scales with candidate count. A publishable result would need a useful improvement in selected-path coverage and correctness at matched cost, or a clear failure characterization. The max-score conformal construction itself is standard.

19. Approach B: a coupled function tube from frozen features

Purpose. Model one shared source of reward error across steps and policies, providing the cleanest initial theoretical route. AdvPO is close prior work [22]; the proposed extension is process coupling and validation.

19.1. A tractable model and an exact induced tube

Start with a frozen feature map $\psi(h) \in \mathbb{R}^d$ and linear real-valued reward $r_w(h) = \psi(h)^\top w$. Fit a regularized head $\hat{w}$ and let

$$V = \lambda I + \sum_{i=1}^n \psi_i \psi_i^\top, \quad \mathcal{C} = \{w : \|w - \hat{w}\|_V \leq \beta_n\}, \quad \mathcal{U} = \{r_w : w \in \mathcal{C}\}.$$

Here $\|u\|_V = \sqrt{u^\top V u}$. The induced pointwise envelope is

$$\psi(h)^\top \hat{w} \pm \beta_n \sqrt{\psi(h)^\top V^{-1} \psi(h)}.$$

Take $\lambda > 0$ so that V is invertible. This is a projection of one ellipsoid; arbitrary simultaneous choices of the pointwise endpoints are not allowed.

For $y_i = \psi_i^\top w^* + \varepsilon_i$, predictable frozen features, conditionally R_ε -sub-Gaussian noise, $\|w^*\| \leq S$, and regularized least squares, a standard confidence choice is

$$\beta_n = R_\varepsilon \sqrt{2 \log \left(\frac{\det(V)^{1/2}}{\det(\lambda I)^{1/2} \delta} \right)} + \sqrt{\lambda} S.$$

Self-normalized theory supplies adaptive-data confidence under these assumptions [31]. This does not justify an unrestricted neural PRM guarantee. Binary conditional means can fit this model only if the linear mean is correctly specified. Labels systematically supplied by a biased judge violate it.

For probability outputs $\text{sigmoid}(\psi^\top w)$, monotonicity yields sigmoid transforms of the logit envelopes *if* the parameter confidence set is valid. Least-squares confidence theory does not automatically validate a logistic head. Hessian/Laplace ellipsoids can be empirical baselines; a formal generalized-linear analysis must establish its own assumptions and radius.

19.2. Joint geometry and robust scoring

For a trajectory feature matrix Ψ_τ whose rows are $\psi(h_t)^\top$, the induced covariance shape is

$$\Sigma_\tau = \Psi_\tau V^{-1} \Psi_\tau^\top.$$

Repeated steps share uncertainty rather than generating new independent errors. For a linear aggregation $a^\top \mathbf{r}$, where a is fixed for the trajectory, the exact worst-case score for the unrestricted linear head is

$$\inf_{w \in \mathcal{C}} a^\top \Psi_\tau w = a^\top \Psi_\tau \hat{w} - \beta_n \sqrt{a^\top \Sigma_\tau a}.$$

This is a support-function identity, not a novel PRM theorem. If rewards must stay in $[0, 1]$, either ensure the ellipsoid satisfies the domain bounds or intersect it with those constraints and solve the resulting constrained problem. Clipping outputs or using sigmoid rewards generally destroys the linear identity. For a minimum objective, $\inf_w \min_t r_w(h_t) = \min_t \inf_w r_w(h_t)$; for products and nonlinear means, simultaneous endpoint substitution need not be exact.

Policy-level pessimism differs from per-trajectory pessimism. With $\bar{\psi}_\pi = \mathbb{E}_\pi \sum_t a_t \psi(h_t)$, the

former equals

$$\bar{\psi}_\pi^\top \hat{w} - \beta_n \|\bar{\psi}_\pi\|_{V^{-1}},$$

while the latter subtracts $\beta_n \mathbb{E}_\pi \|\sum_t a_t \psi(h_t)\|_{V^{-1}}$. Jensen makes the latter more conservative. Report which objective is optimized.

19.3. Misspecification and practical construction

An honest extension is

$$r^*(h) = \psi(h)^\top w^* + m(h), \quad |m(h)| \leq b_{\text{mis}}(h).$$

The resulting set couples parameter error but allows specified residual error. Estimate residual scales on separate problems containing adversarial and on-policy traces. Calling that estimate a uniform bound still requires a coverage argument. A residual floor can protect against zero ensemble variance, but should be evaluated through failures rather than assumed effective.

Implement frozen representations, ridge heads, then independently trained low-rank heads or adapters. Use Cholesky solves rather than explicitly forming V^{-1} . Training needs feature statistics; a full $d \times d$ matrix and quadratic-form scoring can be expensive. Compare dimension reduction, diagonal approximations, and low-rank approximations against full geometry. Do not hide the cost of computing PRM features.

19.4. Research payoff and risks

Compare a shared ellipsoid with its rectangular envelope at matched marginal width and matched joint coverage. Test common-mode bias, repeated steps, and rare logical transformations. The contribution could be a characterization of when coupling avoids excessive conservatism or controls policy exploitation. The main risks are feature misspecification, falsely small radii, and head-only diversity missing shared backbone errors. A generic last-layer ellipsoid is already covered by AdvPO and confidence-set literature.

20. Approach C: structured nuisance and budgeted error tubes

Purpose. Cover systematic shared biases that a seed ensemble misses. This route combines empirical attacks with an interpretable robust set.

Let $g(h)$ describe validated nuisance directions: step duplication, verbosity, unjustified certainty, answer-format markers, or question–solution mismatch. The last is a corruption detector, not a semantics-preserving nuisance. Define

$$r_{u,e}(h) = \hat{r}(h) - g(h)^\top u - e(h), \quad \|u\|_M \leq \rho_b.$$

For a probability target, intersect these constraints with $0 \leq r_{u,e}(h) \leq 1$; the unconstrained penalties below are then conservative bounds and need not be the exact constrained optimum. Alternatively, define the set on real-valued logits and optimize with the explicit link function. One u applies to the full trajectory or policy; specify that scope. Derive g from paired transformations whose labels are checked, not arbitrary text embeddings assumed to represent causality. CoLD motivates the

known-bias case [11]. Put a zero-centered uncertainty set around unexplained effects; an estimated known bias should first correct the center.

For nonnegative aggregation weights, an additional budgeted residual model is

$$|e_t| \leq b_t, \quad \sum_t \frac{|e_t|}{b_t} \leq \Gamma, \quad 0 \leq \Gamma \leq T,$$

omitting zero-width coordinates. Its worst-case linear penalty is the sum of the largest $\lfloor \Gamma \rfloor$ values of $a_t b_t$, plus the fractional part times the next largest. This standard robust-set geometry interpolates between no residual penalty and a rectangular all-steps penalty.

Construction protocol. Create paired correct rewrites and validated logical corruptions on training problems; fit bias features and residual scales; estimate a problem-level deviation budget on a disjoint calibration split; compare an independent box, a shared bias term, and a combined set. If a repeated systematic mistake affects every step, a small Γ is invalid. Use the shared bias component for such common modes and audit the remaining budget assumptions.

Potential contribution. Show that correlated nuisance uncertainty covers known failure directions at smaller useful widths than an unstructured box, then test held-out attacks. Compare against CoLD and hard-negative retraining at equal label cost. Primary risks are unreliable semantic validation and attacks outside g 's span. Token-count penalties alone are a baseline, not the new method.

21. Approach D: a joint reward distribution and robust risk

Purpose. Preserve dependencies in a distributional PRM rather than attach independent distributions to each step. This is an optional extension after simpler tube methods work; scalar distributional pessimism is prior work [27, 25, 28].

Fit a joint predictive distribution $p(\mathbf{r} \mid \tau)$ using shared latent parameters, low-rank factors, or ensemble trajectories. For a specified G , consider a reward-distribution ambiguity set

$$\mathcal{Q}_\rho(\tau) = \{Q : D_{\text{KL}}(Q \| p(\mathbf{r} \mid \tau)) \leq \rho\}.$$

Its robust expected aggregate is

$$\inf_{Q \in \mathcal{Q}_\rho} \mathbb{E}_Q G(\mathbf{r}) = \sup_{\eta > 0} \{-\eta \log \mathbb{E}_p e^{-G(\mathbf{r})/\eta} - \eta \rho\},$$

under the standard duality/integrability conditions, with boundary limits included where needed. The penalized form with fixed η omits $-\eta \rho$; it is a different problem. This variational identity is established robust-risk machinery, not a novel result.

For real-valued Gaussian rewards and linear $G = \mathbf{a}^\top \mathbf{r}$, the penalized score is $\mathbf{a}^\top \boldsymbol{\mu} - (\mathbf{a}^\top \Sigma \mathbf{a}) / (2\eta)$. Correlation changes the penalty; dropping off-diagonal terms can severely underestimate shared error. A Gaussian distribution is not a valid bounded probability distribution unless transformed or truncated, in which case the simple expression need not hold. Monte Carlo log-sum-exp estimates must be checked for tail-sampling error.

Calibration and experiments. Train distributional heads on repeated rollouts in the fixed-completer track or repeated independent annotations in the validity track, accounting for what those

repetitions measure. Compare independent versus joint models, entropic risk versus quantile bounds, and conservative distributions versus the same center without penalties. Measure tail fit after selection and shared errors. Distributional variability alone is not epistemic confidence; a perfectly fitted wrong label mechanism remains wrong. Publishability would require a process-specific gain, not re-deriving scalar entropic rewards.

22. Approach E: optimize, audit, and update the tube

Purpose. Make uncertainty estimation respond to the distribution created by reward optimization. This is the highest-cost combined route.

Define an optimization regime by initial policy, allowed updates, KL/length caps, algorithm, and budget. A reachable family Π_B induces prefix or trajectory distributions, rather than an unspecified ball around the training data. Empirically sample a mixture of checkpoints, best-of- N winners, beam prefixes, and attack-generated traces. A mixture covers its sampled modes; it does not certify every reachable policy.

1. Fit the initial PRM and set on training problems. Freeze its parameters and calibration during a bounded policy-update phase.
2. Optimize the robust score and separately attack it. Select audit examples with high robust score, low ensemble disagreement despite suspicious reasoning, and coverage failures; include random on-policy examples to avoid blind sampling.
3. Obtain independent validity labels or fixed-completer rollout targets. Keep a fresh evaluation stream untouched by reward updates.
4. Retrain the center or uncertainty model, then recalibrate on fresh data. Compare with equal-budget center-only retraining.
5. Validate new policies after the update; use a predefined accept/defer rule when evidence of true improvement or valid coverage is insufficient.

If policy π_k is fixed before fresh calibration and test blocks are drawn under it, a per-round guarantee conditional on the previous history can be combined with $\sum_k \alpha_k \leq \alpha$ through a union bound. This only covers the specified audited deployment blocks. It does not certify policy π_{k+1} trained after that calibration, nor reuse of calibration labels to choose among arbitrary new policies. Uniform function sets or properly justified sequential inference are needed for broader claims.

Weighted calibration is another comparator when labels have a fixed conditional mechanism and reference/on-policy density ratios are available [34]. In text, action-product ratios can be extremely variable. Report effective sample size $(\sum_i w_i)^2 / \sum_i w_i^2$, overlap, and clipping bias. Holding the completer fixed is necessary for the success-probability target; updating the completer changes that target.

Potential contribution. Measure and repair post-optimization coverage loss, showing fewer confidently wrong selected trajectories than static sets, ensembles, and hard-negative training. Main risks are label cost, feedback loops, catastrophic shared verifier mistakes, and moving targets. Treat an adversarial curriculum as empirical adaptation, not automatic uniform robustness.

23. Theoretical characterization: established lemmas and open targets

The propositions below are elementary supporting results derived for these notes. They are not claimed as original publishable theorems. They identify the assumptions a stronger result must supply.

Proposition 1 (Rectangular monotone aggregation). *For a finite trajectory and the independent box $\prod_t [L_t, U_t]$, if G is coordinatewise nondecreasing, then $\inf_{\mathbf{r} \in \prod_t [L_t, U_t]} G(\mathbf{r}) = G(\mathbf{L})$.*

Proof. Every feasible vector is coordinatewise at least $\mathbf{L}$, and $\mathbf{L}$ is feasible. Monotonicity gives both inequalities. For a coupled set $\mathbf{L}$ may not be feasible; its value then remains a conservative lower bound, not an exact robust score. $\square$

Proposition 2 (A simultaneous tube protects a selected candidate). *Suppose $R_j^* \in [L_j, U_j]$ for every member of a finite candidate pool. Let $\hat{j} \in \arg \max_j L_j$, and let $j^* \in \arg \max_j R_j^*$. Then*

$$R_{\hat{j}}^* \geq L_{\hat{j}}, \quad R_{j^*}^* - R_{\hat{j}}^* \leq U_{j^*} - L_{j^*}.$$

Proof. $R_{\hat{j}}^* \geq L_{\hat{j}} \geq L_{j^*}$ and $R_{j^*}^* \leq U_{j^*}$. Subtract. The simultaneous containment event is needed before selection. For half-widths bounded by ϵ , regret is at most 2ϵ . If these are process-aggregate rewards, the bound concerns that aggregate and not automatically final-answer correctness. $\square$

23.1. Why marginal coverage alone fails

Construct a distribution where a fraction α of candidates are wrong, receive score one, and receive an interval that excludes the true reward zero. All other candidates are correct, score 0.8, and are covered. Random-candidate coverage is exactly $1 - \alpha$. Best-of- N chooses an uncovered wrong candidate whenever one occurs, with probability $1 - (1 - \alpha)^N$. At $\alpha = 0.05$, $N = 100$, this is about 0.994. Ties are resolved within wrong candidates. This is a constructed counterexample, not a measured PRM failure. It illustrates why selection-aware theory is necessary even before RL.

Proposition 3 (Robust improvement against a reference). *If the true reward function belongs to a fixed set $\mathcal{U}$, then for every policy*

$$\Delta_{\mathcal{U}}(\pi) := \inf_{r \in \mathcal{U}} \{J_r(\pi) - J_r(\pi_0)\} \leq J_{r^*}(\pi) - J_{r^*}(\pi_0).$$

Therefore $\Delta_{\mathcal{U}}(\pi) > 0$ certifies improvement for that target reward.

Proof. Evaluate the infimum at r^* . The same containment event works for adaptively chosen policies if it is a uniform function-containment event on their domain. $\square$

This reference-centered objective is preferable to comparing two separate pessimistic values: improvement in $\inf_r J_r(\pi)$ does not alone prove improvement in the true return. Approximate occupancy estimates, optimization error, and uncertain labels each require an explicit error allowance.

23.2. A shift-aware bound with transparent assumptions

Suppose a bounded trajectory utility satisfies $R^* = \hat{R} + e$ and $\mathbb{E}_{P_0} e^2 \leq \epsilon^2$, with $P_\pi \ll P_0$. For $w = dP_\pi/dP_0$, Cauchy–Schwarz gives

$$\left| \mathbb{E}_{P_\pi} R^* - \mathbb{E}_{P_\pi} \hat{R} \right| \leq \epsilon \sqrt{\mathbb{E}_{P_0} w^2} = \epsilon \sqrt{1 + \chi^2(P_\pi \| P_0)}.$$

For improvement relative to P_0 , the error difference is bounded by $\epsilon \sqrt{\chi^2(P_\pi \| P_0)}$, using $w - 1$. This is a standard change-of-measure calculation related to correlated-proxy regularization [18]; it fails when relevant support is absent or the ratio is unbounded. Replacing whole trajectories by step occupancy needs consistent normalization and an additive reward objective.

23.3. Open theoretical targets

1. **Geometry versus conservatism:** characterize regret or safe improvement for boxes versus shared ellipsoids at equal joint coverage, including correlated misspecification and variable-length reasoning. A dual-norm identity alone is insufficient novelty.
2. **Coverage under specified optimization:** prove selected-trajectory coverage for a feasible protocol beyond naive NT union bounds, explicitly comparing with JOMI and problem-block calibration. Specify the target and budget.
3. **Necessity and lower bounds:** construct families where any narrow set calibrated only on a reference policy misses a reachable adversarial mode; state the needed overlap or complexity assumptions for informative protection.
4. **Repair-aware process uncertainty:** characterize a set over validity transitions and show how robust outcome estimation differs from weakest-step scoring, comparing with RSP. This is more ambitious and needs reliable labels.

A convergence theorem for a neural min–max method should not be asserted from convex-head or strongly monotone toy assumptions. Prove the restricted model first, and state the empirical extension separately.

24. An experiment plan that can support a paper

24.1. Hypotheses and staged scope

- H1:** Static pointwise coverage degrades among PRM-selected candidates as search pressure increases.
- H2:** A calibrated tube reduces false high-reward selections better than equal-cost pessimism, label improvement, or reduced search.
- H3:** Coupling improves the coverage–sharpness–accuracy tradeoff when errors share structure, but can fail under unmodeled common bias.
- H4:** Static calibration degrades under policy movement; independently audited updates recover useful coverage in specified regimes.

Begin with frozen-pool inference and a synthetic setting with known truth. Proceed to RL only if uncertainty is informative and an overoptimization baseline can be reproduced. A pure empirical paper can stop at inference-time claims; it should not imply mitigation of RL hacking without RL experiments.

24.2. Data and split protocol

Use PRM800K or appropriate process training data for heads and adapters, with training, validation, calibration, and test splits separated by problem. All paraphrases, injected errors, sampled solutions, and prefixes of a problem belong to one split. Use ProcessBench, PRMBench, and Socratic-PRMBench as independent diagnostic benchmarks; do not train a calibrator on their final evaluation labels.

For answer evaluation, generate trajectories on held-out MATH-family problems, with GSM-style tasks as a second difficulty regime. Small AIME sets are a stress test with wide uncertainty, not the sole source of statistical claims. If a published baseline calibrates on MATH500, make that data use explicit and evaluate on a distinct held-out set. Check upstream training contamination where possible, and qualify residual uncertainty.

Use at least two generator sizes or families and at least two PRM families, such as Qwen2.5-Math-PRM and Skywork-o1 PRMs as documented in the cited work. These are reproducibility candidates, not claims about the latest model releases. Verify model revisions and inference interfaces before running experiments.

24.3. Baselines and ablations

Comparator	Confound it addresses
Nominal single PRM; calibrated center	Is the benefit better point predictions?
Mean, minimum, and variance-penalized ensembles	Standard reward pessimism
AdvPO-style frozen-head confidence score	Existing confidence-region optimization
Quantile PRM; conditional-OT calibration	Existing prefix uncertainty modeling
Caution novelty penalty	Single-model OOD conservatism
PURE-style minimum; sum/mean/product controls	Credit assignment versus uncertainty
Reduced N , early stopping, and KL sweep	Less optimization versus a better reward
Length correction and hard-negative retraining	Bias correction and improved labels
Box versus coupled tube; calibrated versus tuned radius	Geometry and calibration
Pointwise, block, union-bound, selection-conditional variants	What survives selection

Comparator	Confound it addresses
Random widths; constant widths; oracle error widths	Whether width identifies errors
Reliable outcome-only RLVR; hybrid rewards	Whether a PRM improves a reliable task signal

A full factorial is expensive. Predefine a core set containing nominal, calibrated-center, ensemble pessimism, AdvPO, one PRM calibration method, Caution, and the proposed tube; use targeted secondary ablations. Match center quality, training labels, total generation/scoring cost, and where relevant measured KL. Radius must be calibrated on calibration data or tuned on validation, never on the test set. Report those variants separately.

24.4. Optimization stress tests

For frozen candidates, sweep $N \in \{1, 4, 16, 64, 256\}$, subject to a common token budget, scoring nested candidate pools. Report both equal- N and equal-compute comparisons. Test step splitting/merging, repetition, neutral padding, incorrect numeric transformations, unsupported premises, prompt mismatch, suffix attacks, and delimiter manipulation. Validate whether each edit preserves or changes correctness, and keep an untouched transfer family.

For RL, freeze a reward model and tube during each comparison run. Use equal initial policies, prompts, token caps, training compute, and KL coefficients; add a matched-observed-KL analysis because penalties can change policy movement. The primary controlled comparison assigns the same trajectory aggregation to nominal and robust rewards. Compare process-only and hybrid-outcome supervision separately. A maximum-empty-response score is a parser defect to fix and report, not a gain to accept. Save checkpoints and raw rewards before normalization.

Attack the deployed robust score, including its uncertainty head. A defense that works only against attacks optimized for $\hat{r}$ may simply move the exploit to $\hat{r} - b$. Black-box search and white-box token attacks are separate budgets; report access assumptions, calls, failed attempts, and transfer results.

24.5. Metrics and statistical reporting

The primary metric is held-out answer correctness under increasing pressure, supplemented by independent process validity on a stratified audit subset. Use answer checkers with canonicalization and spot checks; do not call a larger LLM an exact oracle. Score trajectories with correct answers but flawed processes as a separate category.

Report step, trajectory, candidate-pool, and selected-trajectory coverage; interval width and singleton/abstention rates; false acceptance of invalid steps; overestimation error among top-scoring candidates; reward-accuracy curves; length, repetition, vacuity, KL, and compute. For bounded aligned utilities, also report lower-bound violations $(L - R^*)_+$ and their tail severity. Do not subtract an uncalibrated free scalar reward from a binary accuracy metric and interpret the result as a meaningful numerical “hacking gap.”

Use paired candidate pools, multiple independent head/policy seeds, and problem-level bootstrap confidence intervals. A starting plan is at least three seeds for expensive RL and five for

lightweight methods; justify additional runs from variance and a pilot power analysis. Resampling individual steps as if independent understates uncertainty. Report all prespecified pressure levels, including unsuccessful conditions. Specify checkpoints by validation or fixed training budget, never retrospective test accuracy.

24.6. Synthetic experiments with known ground truth

Build finite reasoning trees with known validity labels and optional repair transitions. Expose features containing a true validity signal and spurious style dimensions. Vary reference-policy coverage, shared versus independent reward error, trajectory length, repair rate, and optimization budget. Compare nominal search/RL, boxes, ellipsoids, bias sets, and oracle bounds. This identifies when the set really contains truth and when apparent conservatism only shrinks optimization. Synthetic results validate assumptions, not realistic LLM claims.

25. Recommended route and publication decision gates

Route	Minimum convincing result	Main limitation
Empirical	Selected-path coverage failures plus a useful, equal-cost repair on multiple PRMs and pressure regimes	Prediction sets may be too wide; no RL claim without RL evidence
Theoretical	A process-specific coverage/regret or impossibility result beyond standard confidence-set lemmas	Strong feature/support assumptions may limit practical implications
Combined	A restricted theorem, informative real-PRM tubes, and adaptation-aware stress tests that match the theorem’s mechanism	Highest label and training cost

Recommended starting point: Approach A as a diagnostic and calibrated baseline, Approach B as the structured candidate method, and synthetic tests to separate shared uncertainty from pointwise errors. Add a misspecification term from Approach C only when common-mode failures are observed. Approach E is the RL extension; Approach D is a later alternative, not a prerequisite.

1. **Target gate:** a label and reward definition must be coherent. If success-probability bounds are the only informative ones, narrow the claim to that target instead of calling them correctness bounds.
2. **Information gate:** uncertainty must predict overestimation or useful abstention better than constant/random widths. Otherwise improve the representation or report why this uncertainty signal fails.
3. **Mechanism gate:** show gains at fixed aggregation and comparable compute/optimization pressure. If reduced search explains the effect, do not claim a stronger reward estimator.

4. **Prior-work gate:** compare explicitly with AdvPO, PRM quantile/OT calibration, ensemble pessimism, Caution, and selection-aware inference. Update the novelty search before drafting a submission.
5. **Robustness gate:** test adaptive attacks and held-out families. If calibration fails under RL, retain an honest inference-only result or study the failure as the main contribution.

The immediate deliverable is a research design, not an experiment result or a guarantee of publication. A well-supported negative finding about tube coverage may also be valuable, provided it is broad, controlled, and explains a mechanism that existing uncertainty methods miss.

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